Introduction to Database

Semester Final Project

Pakistan Census Database

Session: Spring 2026

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1. Project Title

Pakistan Census Database

2. Project Overview

The **Pakistan Census Database** is a centralized system built to collect and manage national demographic, household, and geographic data.

- **Geographic Tracking:** Organizes data down a strict chain from Provinces and Districts to local Census Blocks.
- **Property & Housing Utilities:** Tracks the quality of physical structures and access to water, power, and fuel.
- **Population Registries:** Stores citizen profiles, including CNICs, family structures, literacy, and income.
- **Field Operations:** Manages the census workforce by linking local enumerators to their regional supervisors.

Ultimately, this system provides fast, reliable data slicing to help the government make accurate decisions regarding resource allocation and national planning.

3. Objectives

Objective 1: Securely store and quickly find data across all geographic levels, from whole provinces down to local census blocks.

Objective 2: Track citizen details accurately, including CNICs, family structures, income, literacy, and housing conditions.

Objective 3: Support generating clear population reports, literacy rate summaries, and regional development statistics for government planning.

Objective 4: Perform complex SQL queries to filter and analyze deep demographic trends and employment rates across different regions.

Objective 5: Ensure database normalization and strict integrity constraints so that data remains clean, connected, and free of duplicates.

4. Scope

What is Included:

- **Geographic & Personnel Infrastructure:** Tracking administrative boundaries from provinces down to local census blocks, along with the operational field staff hierarchies.
- **Property & Household Metrics:** Managing details of physical structures, domestic utility profiles, and family household groupings.
- **Citizen Demographics:** Storing individual citizen records, including identity verification numbers (CNICs), family relations, educational status, and income levels.
- **Backend Database Optimization:** Developing index structures, relational constraints, and core SQL queries for backend data analysis.

What is Not Included:

- **Front-End Applications:** The project does not include building web or mobile applications for data entry or public dashboards.
- **Live GPS Tracking:** It maps static coordinates for structures but does not support real-time GPS tracking of field staff.
- **Biometric Verification System:** The database processes text-based CNIC records but does not handle direct fingerprint or facial biometric verification processing.

5. Entities and Attributes

Entity	Attributes
Water_Source	**water_source_id**, source_name
Lighting_Source	**lighting_source_id**, source_name
Cooking_Fuel_Source	**fuel_source_id**, fuel_source_name
Washroom_Status	**washroom_status_id**, status_name
Kitchen_Status	**kitchen_status_id**, status_name
Biological_Sex	**sex_id**, sex_name
Religion	**religion_id**, religion_name
Mother_Tongue	**mother_tongue_id**, tongue_name
Education_Level	**education_level_id**, level_name
Employment_Status	**employment_status_id**, status_name
Marital_Status	**marital_status_id**, status_name
Disability_Type	**disability_type_id**, disability_name
Occupation	**occupation_id**, occupation_name
Industry	**industry_id**, industry_name
Relationship_Type	**relationship_type_id**, relationship_name

Entity	Attributes
Census_Cycle	**census_id**, census_year, census_title, start_date, end_date
Census_Staff	**staff_id**, first_name, last_name, staff_role, contact_number, email, assigned_census_block_id, supervisor_id, employment_date

Entity	Attributes
Nationality	**nationality_id**, country_name, citizenship_status
Province	**province_id**, province_name
Division	**division_id**, division_name, province_id
District	**district_id**, district_name, division_id
Tehsil	**tehsil_id**, tehsil_name, district_id
Union_Council	**union_council_id**, union_council_name, union_council_classification_id, tehsil_id
Census_Block	**census_block_id**, census_block_code, estimated_household_count, is_urban, area_sq_km, union_council_id
Union_Council_Classification	**union_council_classification_id**, classification_name, description

Entity	Attributes
Person	**person_id**, household_id, first_name, last_name, cnic_number, sex_id, date_of_birth, nationality_id, religion_id, mother_tongue_id, marital_status_id, relationship_type_id, head_person_id, is_literate, education_level_id, currently_attending_school, employment_status_id, occupation_id, industry_id, monthly_income, data_entry_operator_id, created_date
Person_Disability	**person_disability_id**, **person_id**, disability_type_id
Migration_History	**migration_id**, **person_id**, previous_district_id, migration_reason, year_of_migration

Entity	Attributes
Structure_Type	**structure_type_id**, type_name, description
Structure	**structure_id**, structure_address_no, gps_latitude, gps_longitude, created_date, census_block_id, structure_type_id, water_source_id, lighting_source_id, fuel_source_id, washroom_status_id, kitchen_status_id
Structure_Residential	**structure_id**, total_residential_units
Structure_Commercial	**structure_id**, commercial_type_id, business_name, estimated_active_employees, has_industrial_power_connection, has_active_license
Structure_Institution	**structure_id**, institution_type, organization_name, population_count, is_government
Commercial_Sector_Type	**commercial_type_id**, commercial_type_name, description
Household	**household_id**, structure_id, census_id, household_sub_token, household_category_id, residential_status_type_id, owner_sex_id, number_of_rooms, household_size, data_entry_operator_id, created_date
Household_Category	**household_category_id**, category_name, description
Residential_Status_Type	**residential_status_type_id**, status_name, description

6. Relationships

1. Geographic Hierarchy Chain

Province to Division:

- ***Cardinality:*** **1:M** (One Province can have Many Divisions).
- ***Participation:*** **Total** on Division side (A division cannot exist without being mapped to a province).
- ***Foreign Key:*** Division.province_id references Province.province_id.

Division to District:

- ***Cardinality:*** **1:M** (One Division contains Many Districts).
- ***Participation:*** **Total** on District side (A district must belong to an administrative division).
- ***Foreign Key:*** District.division_id references Division.division_id.

District to Tehsil:

- ***Cardinality:*** **1:M** (One District oversees Many Tehsils).
- ***Participation:*** **Total** on Tehsil side.
- ***Foreign Key:*** Tehsil.district_id references District.district_id.

Tehsil to Union Council:

- ***Cardinality:*** **1:M** (One Tehsil contains Many Union Councils).
- ***Participation:*** **Total** on Union Council side.
- ***Foreign Key:*** Union_Council.tehsil_id references Tehsil.tehsil_id.

Union Council to Census Block:

- ***Cardinality*:** **1:M** (One Union Council contains Many localized Census Blocks).
- ***Participation*:** **Total** on Census Block side.
- ***Foreign Key*:** Census_Block.union_council_id references Union_Council.union_council_id.

2. Building Extension Layers

Structure to Structure_Residential:

- ***Cardinality*:** **1:1**
- ***Participation*:** **Partial** on Structure side, **Total** on Residential side (The profile cannot exist without its base parent building asset).
- ***Foreign Key*:** Structure_Residential.structure_id references Structure.structure_id.

Structure to Structure_Commercial:

- ***Cardinality*:** **1:1**
- ***Participation*:** **Partial** on Structure side, **Total** on Commercial side.
- ***Foreign Key*:** Structure_Commercial.structure_id references Structure.structure_id.

Structure to Structure_Institution:

- ***Cardinality*:** **1:1**.
- ***Participation*:** **Partial** on Structure side, **Total** on Institution side.
- ***Foreign Key*:** Structure_Institution.structure_id references Structure.structure_id.

3. Core Operational Census Operations

Census_Block to Structure Location:

- ***Cardinality:*** **1:M** (One Census Block encompasses Many physical building structures).
- ***Participation:*** **Total** on Structure side (Every building must be cataloged inside an active census block boundary).
- ***Foreign Key:*** Structure.census_block_id references Census_Block.census_block_id.

Structure Location to Household Unit:

- ***Cardinality:*** **1:M** (A physical building structure can host Multiple separate living households).
- ***Participation:*** **Total** on Household side (Every household must be linked to a physical address/structure asset).
- ***Foreign Key:*** Household.structure_id references Structure.structure_id.

Household Unit to Person Profile:

- ***Cardinality:*** **1:M** (One Household unit contains Many citizens living/eating together).
- ***Participation:*** **Total** on Person side (An individual citizen must be accounted for within an registered household cohort).
- ***Foreign Key:*** Person.household_id references Household.household_id.

4. Recursive Self-Referencing Links

Census_Staff to Census_Staff (Management Hierarchy):

- ***Cardinality*:** **1:M** (One Supervisor manages Many field staff members/enumerators).
- ***Participation*:** **Total** on enumerators side (Enumerators have supervisors).
- ***Foreign Key*:** Census_Staff.supervisor_id references Census_Staff.staff_id.

Person to Person (Household Head Mapping):

- ***Cardinality*:** **1:M** (One household head is referenced by Many living family members in the house).
- ***Participation*:** **Partial** on both sides (A chosen household head points to null or themselves, while family members point to the head).
- ***Foreign Key*:** Person.head_person_id references Person.person_id.

5. Junction Tables

Person to Disability_Type via Person_Disability:

- ***Cardinality*:** **M:N** (One person can have multiple concurrent functional health challenges; one disability type applies to many citizens).
- ***Participation*:** **Total** on the junction table (Person_Disability), **Partial** on the individual tables (A citizen might have zero disabilities listed).
- ***Foreign Keys:*** Person_Disability.person_id references Person.person_id. Person_Disability.disability_type_id references Disability_Type.disability_type_id.

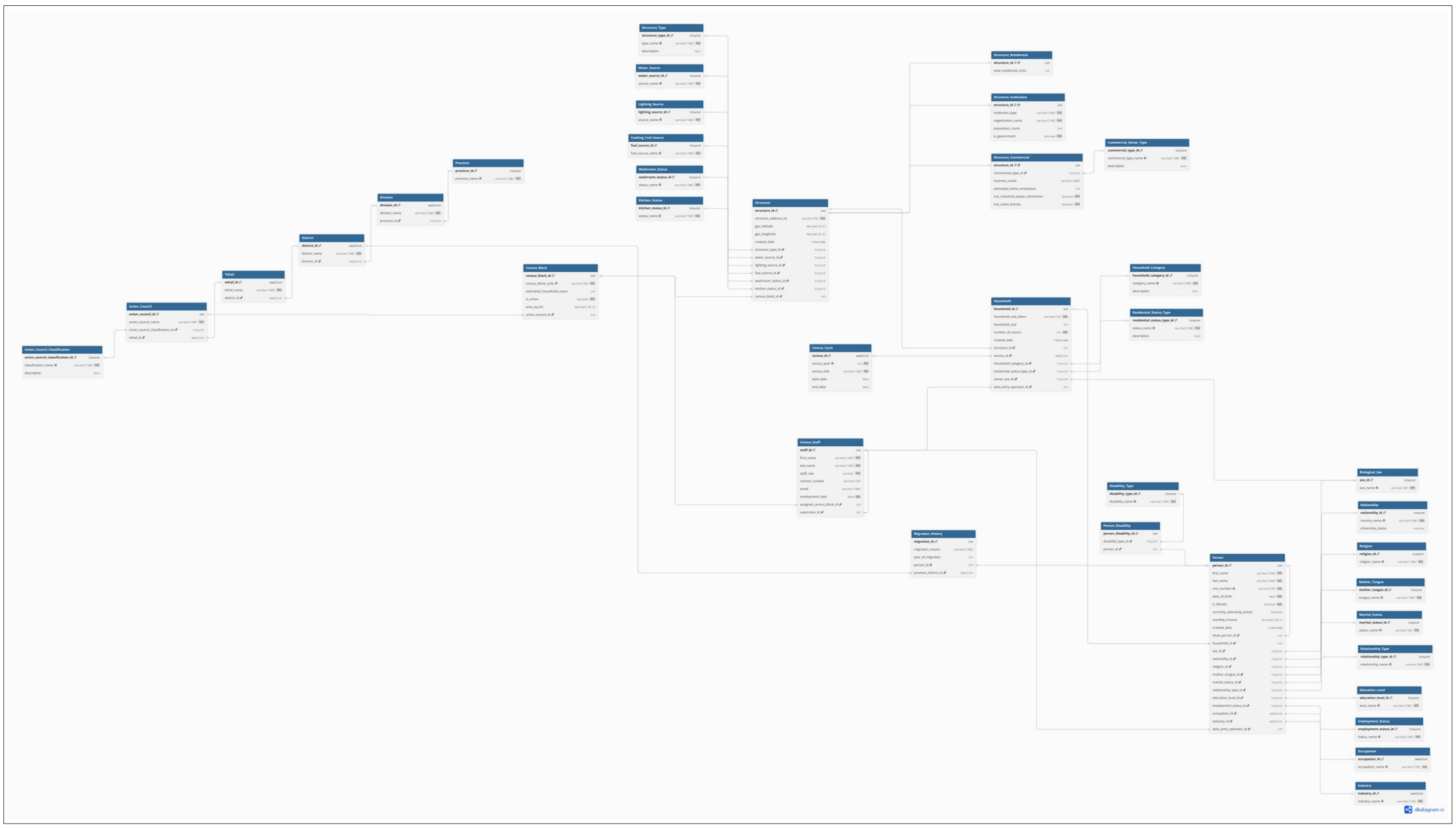
Person to District via Migration_History:

- ***Cardinality:*** **M:N** (A citizen can move through multiple locations over time; a single district acts as a historical home for many citizens).

- ***Participation:*** **Total** on the history tracker (Migration_History), **Partial** on individual tables.

- ***Foreign Keys:*** Migration_History.person_id references Person.person_id. Migration_History.previous_district_id references District.district_id.

7. ERD (Entity Relationship Diagram)



8. Relational Schema*

- **Water_Source (water_source_id, source_name)**
- **Lighting_Source** (**lighting_source_id**, source_name)
- **Cooking_Fuel_Source** (**fuel_source_id**, fuel_source_name)
- **Washroom_Status** (**washroom_status_id**, status_name)
- **Kitchen_Status** (**kitchen_status_id**, status_name)
- **Household_Category** (**household_category_id**, category_name, description)
- **Residential_Status_Type** (**residential_status_type_id**, status_name, description)
- **Biological_Sex** (**sex_id**, sex_name)
- **Religion** (**religion_id**, religion_name)
- **Mother_Tongue** (**mother_tongue_id**, tongue_name)
- **Education_Level** (**education_level_id**, level_name)
- **Employment_Status** (**employment_status_id**, status_name)
- **Marital_Status** (**marital_status_id**, status_name)
- **Disability_Type** (**disability_type_id**, disability_name)
- **Occupation** (**occupation_id**, occupation_name)
- **Industry** (**industry_id**, industry_name)
- **Relationship_Type** (**relationship_type_id**, relationship_name)
- **Nationality** (**nationality_id**, country_name, citizenship_status)
- **Union_Council_Classification** (**union_council_classification_id**, classification_name, description)

- **Census_Cycle** (**census_id**, census_year, census_title, start_date, end_date)
- **Structure_Type** (**structure_type_id**, type_name, description)
- **Commercial_Sector_Type** (**commercial_type_id**, commercial_type_name, description)
- **Province** (**province_id**, province_name)
- **Division** (**division_id**, division_name, *province_id references Province.province_id*)
- **District** (**district_id**, district_name, *division_id references Division.division_id*)
- **Tehsil** (**tehsil_id**, tehsil_name, *district_id references District.district_id*)
- **Union_Council** (**union_council_id**, union_council_name, *union_council_classification_id references Union_Council_Classification.union_council_classification_id*, *tehsil_id references Tehsil.tehsil_id*)
- **Census_Block** (**census_block_id**, census_block_code, estimated_household_count, is_urban, area_sq_km, *union_council_id references Union_Council.union_council_id*)
- **Census_Staff** (**staff_id**, first_name, last_name, staff_role, contact_number, email, *assigned_census_block_id references Census_Block.census_block_id*, *supervisor_id references Census_Staff.staff_id*, employment_date)
- **Structure** (**structure_id**, structure_address_no, gps_latitude, gps_longitude, created_date, *census_block_id references Census_Block.census_block_id*, *structure_type_id references Structure_Type.structure_type_id*, *water_source_id references Water_Source.water_source_id*, *lighting_source_id references Lighting_Source.lighting_source_id*, *fuel_source_id references Cooking_Fuel_Source.fuel_source_id*, *washroom_status_id references Washroom_Status.washroom_status_id*, *kitchen_status_id references Kitchen_Status.kitchen_status_id*)
- **Structure_Institution** (***structure_id*** *references Structure.structure_id*, institution_type, organization_name, population_count, is_government)

- **Structure_Commercial** (***structure_id*** *references Structure.structure_id*, *commercial_type_id references Commercial_Sector_Type.commercial_type_id*, business_name, estimated_active_employees, has_industrial_power_connection, has_active_license)

- **Structure_Residential** (***structure_id*** *references Structure.structure_id*, total_residential_units)

- **Household** (**household_id**, *structure_id references Structure.structure_id*, *census_id references Census_Cycle.census_id*, household_sub_token, *household_category_id references Household_Category.household_category_id*, *residential_status_type_id references Residential_Status_Type.residential_status_type_id*, *owner_sex_id references Biological_Sex.sex_id*, number_of_rooms, household_size, *data_entry_operator_id references Census_Staff.staff_id*, created_date)

- **Person** (**person_id**, *household_id references Household.household_id*, first_name, last_name, cnic_number, *sex_id references Biological_Sex.sex_id*, date_of_birth, *nationality_id references Nationality.nationality_id*, *religion_id references Religion.religion_id*, *mother_tongue_id references Mother_Tongue.mother_tongue_id*, *marital_status_id references Marital_Status.marital_status_id*, *relationship_type_id references Relationship_Type.relationship_type_id*, *head_person_id references Person.person_id*, is_literate, *education_level_id references Education_Level.education_level_id*, currently_attending_school, *employment_status_id references Employment_Status.employment_status_id*, *occupation_id references Occupation.occupation_id*, *industry_id references Industry.industry_id*, monthly_income, *data_entry_operator_id references Census_Staff.staff_id*, created_date)

- **Person_Disability** (**person_disability_id**, *person_id references Person.person_id*, *disability_type_id references Disability_Type.disability_type_id*)

- **Migration_History** (**migration_id**, *person_id references Person.person_id*, *previous_district_id references District.district_id*, migration_reason, year_of_migration)

*HOW TO READ

- Fields styled in **bold** represent the **Primary Key (PK)** of the table.

- Fields styled in *italics* represent a **Foreign Key (FK)** along with its explicit relational table direction map.

- Fields styled in ***bold italics*** represent a composite setup acting concurrently as both the table's Primary Key and its parent class Foreign Key identifier (1:1 structural extension).

9. SQL Queries and Data Retrieval

A. Joins

INNER JOIN

Shows top 5 districts in Pakistan with highest number of households.

```
mysql> SELECT
    ->     p.province_name,
    ->     d.district_name,
    ->     SUM(cb.estimated_household_count) AS total_population
    -> FROM Province p
    -> INNER JOIN Division dv ON p.province_id = dv.province_id
    -> INNER JOIN District d ON dv.division_id = d.division_id
    -> INNER JOIN Tehsil t ON d.district_id = t.district_id
    -> INNER JOIN Union_Council uc ON t.tehsil_id = uc.tehsil_id
    -> INNER JOIN Census_Block cb ON uc.union_council_id = cb.union_council_id
    -> GROUP BY p.province_name, d.district_name
    -> ORDER BY Total_Population DESC
    -> LIMIT 5;
+------------------------------+-----------------+-------------------+
| province_name                | district_name   | total_population  |
+------------------------------+-----------------+-------------------+
| Sindh                        | Karachi Central |              3440 |
| Sindh                        | Karachi South   |              3350 |
| Islamabad Capital Territory  | Islamabad Urban |              2800 |
| Sindh                        | Karachi East    |              2650 |
| Sindh                        | Hyderabad       |              1910 |
+------------------------------+-----------------+-------------------+
5 rows in set (0.01 sec)
```

Shows people who have Bachelors, Masters or PhD degree with their job and income.

```
mysql> SELECT
    ->     CONCAT(p.first_name, ' ', p.last_name) AS name,
    ->     el.level_name,
    ->     es.status_name,
    ->     o.occupation_name,
    ->     i.industry_name,
    ->     p.monthly_income
    -> FROM Person p
    -> INNER JOIN Education_Level el ON p.education_level_id = el.education_level_id
    -> INNER JOIN Employment_Status es ON p.employment_status_id = es.employment_status_id
    -> INNER JOIN Occupation o ON p.occupation_id = o.occupation_id
    -> INNER JOIN Industry i ON p.industry_id = i.industry_id
    -> WHERE UPPER(el.level_name) IN ('BACHELORS', 'MASTERS', 'PHD');
+---------------+------------+--------------------+------------------------------+--------------------------------+----------------+
| name          | level_name | status_name        | occupation_name              | industry_name                  | monthly_income |
+---------------+------------+--------------------+------------------------------+--------------------------------+----------------+
| Muhammad Khan | Masters    | Employed Full-Time | Doctor/Medical Professional  | Public Administration          |       45000.00 |
| Ali Khan      | Masters    | Student            | Government Official          | Wholesale and Retail Trade     |           0.00 |
| Malik Hassan  | PHD        | Employed Full-Time | Primary School Teacher       | Utilities (Water/Electricity)  |       35000.00 |
| Khalid Ahmed  | Bachelors  | Employed Full-Time | Tailor/Seamstress            | Transportation and Logistics   |       55000.00 |
+---------------+------------+--------------------+------------------------------+--------------------------------+----------------+
4 rows in set (0.00 sec)
```

LEFT JOIN

Shows all persons with their education, job, occupation and nationality — even if some info is missing.

```
mysql> SELECT
    ->     CONCAT(p.first_name, ' ', p.last_name) AS name,
    ->     el.level_name,
    ->     es.status_name AS employment_status,
    ->     o.occupation_name,
    ->     i.industry_name,
    ->     n.country_name
    -> FROM Person p
    -> LEFT JOIN Education_Level el ON p.education_level_id = el.education_level_id
    -> LEFT JOIN Employment_Status es ON p.employment_status_id = es.employment_status_id
    -> LEFT JOIN Occupation o ON p.occupation_id = o.occupation_id
    -> LEFT JOIN Industry i ON p.industry_id = i.industry_id
    -> LEFT JOIN Nationality n ON p.nationality_id = n.nationality_id
    -> ORDER BY el.level_name;
+----------------+--------------+--------------------------------+------------------------------+--------------------------------+--------------+
| name           | level_name   | employment_status              | occupation_name              | industry_name                  | country_name |
+----------------+--------------+--------------------------------+------------------------------+--------------------------------+--------------+
| Khalid Ahmed   | Bachelors    | Employed Full-Time             | Tailor/Seamstress            | Transportation and Logistics   | Pakistan     |
| Hamza Hassan   | Intermediate | Employed Full-Time             | Carpenter                    | Telecommunications             | Pakistan     |
| Danial Khan    | Intermediate | Employed Full-Time             | Construction Laborer         | Public Administration          | Pakistan     |
| Ayesha Khan    | Intermediate | Student                        | NULL                         | NULL                           | Pakistan     |
| Kamil Ahmad    | Intermediate | Employed Full-Time             | Plumber                      | Manufacturing - Other          | Pakistan     |
| Taufiq Malik   | Intermediate | Employed Full-Time             | Engineer                     | Transportation and Logistics   | Pakistan     |
| Rohan Hassan   | Intermediate | Employed Full-Time             | Electrician                  | Public Administration          | Pakistan     |
| Karim Ali      | Intermediate | Employed Full-Time             | Shopkeeper/Retail Merchant   | Health and Social Work         | Pakistan     |
| Mir Khan       | Intermediate | Employed Full-Time             | Nurse                        | Construction                   | Pakistan     |
| Farhan Hassan  | Intermediate | Employed Full-Time             | Driver/Transport Worker      | Education                      | Pakistan     |
| Dina Ahmed     | Intermediate | Employed Full-Time             | NULL                         | NULL                           | Pakistan     |
| Bashir Khan    | Intermediate | Student                        | Engineer                     | Wholesale and Retail Trade     | Pakistan     |
| Abbas Malik    | Intermediate | Employed Full-Time             | Tailor/Seamstress            | Manufacturing - Textiles       | Pakistan     |
| Muhammad Khan  | Masters      | Employed Full-Time             | Doctor/Medical Professional  | Public Administration          | Pakistan     |
| Ali Khan       | Masters      | Student                        | Government Official          | Wholesale and Retail Trade     | Pakistan     |
| Imad Ali       | Matriculation | Student                       | NULL                         | NULL                           | Pakistan     |
| Ahmed Ali      | Matriculation | Employed Full-Time            | Driver/Transport Worker      | Agriculture and Farming        | Pakistan     |
| Arjun Ali      | Matriculation | Student                       | NULL                         | NULL                           | Pakistan     |
| Akhtar Hassan  | Matriculation | Employed Full-Time            | Business Owner               | Hospitality and Tourism        | Pakistan     |
| Mansoor Hassan | Matriculation | Employed Full-Time            | Driver/Transport Worker      | Public Administration          | Pakistan     |
| Zubair Hassan  | Matriculation | Employed Full-Time            | Factory Worker               | Health and Social Work         | Pakistan     |
| Wasim Malik    | Matriculation | Employed Full-Time            | Agricultural Farmer          | Manufacturing - Textiles       | Pakistan     |
| Rana Ahmed     | Matriculation | Employed Full-Time            | Domestic Helper/Maid         | Health and Social Work         | Pakistan     |
| Manzoor Ahmad  | Matriculation | Employed Full-Time            | Construction Laborer         | Wholesale and Retail Trade     | Pakistan     |
| Farah Khan     | Matriculation | Employed Full-Time            | NULL                         | NULL                           | Pakistan     |
| Nadim Khan     | Matriculation | Employed Full-Time            | Government Official          | Arts and Entertainment         | Pakistan     |
| Rasheed Ali    | Matriculation | Employed Full-Time            | Nurse                        | Wholesale and Retail Trade     | Pakistan     |
```

Shows all married persons with their household size information.

```
mysql> SELECT
    ->     CONCAT(p.first_name, ' ', p.last_name) AS name,
    ->     ms.status_name,
    ->     h.household_size
    -> FROM Person p
    -> LEFT JOIN Marital_Status ms ON p.marital_status_id = ms.marital_status_id
    -> LEFT JOIN Household h ON p.household_id = h.household_id
    -> WHERE ms.status_name = 'Married';
+---------------+-------------+----------------+
| name          | status_name | household_size |
+---------------+-------------+----------------+
| Muhammad Khan | Married     |              5 |
| Fatima Khan   | Married     |              5 |
| Ahmed Ali     | Married     |              4 |
| Noor Ali      | Married     |              4 |
| Rashid Hassan | Married     |              4 |
| Zainab Hassan | Married     |              4 |
| Nasir Hussain | Married     |              4 |
| Amina Hussain | Married     |              4 |
| Samir Khan    | Married     |              3 |
| Maha Khan     | Married     |              3 |
| Waqar Ahmed   | Married     |              4 |
| Rania Ahmed   | Married     |              4 |
| Jamal Malik   | Married     |              3 |
| Lia Malik     | Married     |              3 |
| Tariq Hassan  | Married     |              3 |
| Nadia Hassan  | Married     |              3 |
| Yousef Ali    | Married     |              3 |
| Hana Ali      | Married     |              3 |
| Saad Khan     | Married     |              3 |
| Luna Khan     | Married     |              3 |
| Adnan Ahmed   | Married     |              3 |
| Yasmin Ahmed  | Married     |              3 |
| Wasim Malik   | Married     |              3 |
| Rena Malik    | Married     |              3 |
| Mohan Hassan  | Married     |              3 |
| Anjali Hassan | Married     |              3 |
| Deepak Ali    | Married     |              3 |
| Priya Ali     | Married     |              3 |
| Siddiqui Khan | Married     |              3 |
| Gulnar Khan   | Married     |              3 |
| Zahir Ahmad   | Married     |              3 |
| Meena Ahmad   | Married     |              3 |
| Ibrahim Raza  | Married     |              3 |
| Saira Raza    | Married     |              3 |
```

RIGHT JOIN

Shows urban census blocks with more than 300 households and their assigned staff.

```
mysql> SELECT
    ->     CONCAT(cs.first_name, ' ', cs.last_name) AS staff_name,
    ->     cs.staff_role,
    ->     cs.contact_number,
    ->     cb.census_block_code,
    ->     cb.estimated_household_count,
    ->     cb.area_sq_km,
    ->     uc.union_council_name,
    ->     t.tehsil_name,
    ->     d.district_name
    -> FROM Census_Staff cs
    -> RIGHT JOIN Census_Block cb ON cs.assigned_census_block_id = cb.census_block_id
    -> RIGHT JOIN Union_Council uc ON cb.union_council_id = uc.union_council_id
    -> RIGHT JOIN Tehsil t ON uc.tehsil_id = t.tehsil_id
    -> RIGHT JOIN District d ON t.district_id = d.district_id
    -> WHERE cb.is_urban = TRUE AND cb.estimated_household_count > 300 AND (cs.staff_role IN ('Enumerator', 'Supervisor') OR cs.staff_role IS NULL)
    -> ORDER BY cb.estimated_household_count DESC;
```

staff_name	staff_role	contact_number	census_block_code	estimated_household_count	area_sq_km	union_council_name	tehsil_name	district_name
NULL	NULL	NULL	BLK-UC062	950	2.10	UC Lyari	Saddar (KHI South)	Karachi South
NULL	NULL	NULL	BLK-UC070	920	1.70	UC Azizabad	Gulberg (KHI)	Karachi Central
NULL	NULL	NULL	BLK-UC069	880	1.90	UC Federal B Area	Gulberg (KHI)	Karachi Central
NULL	NULL	NULL	BLK-UC064	850	1.50	UC Kharadar	Aram Bagh	Karachi South
NULL	NULL	NULL	BLK-UC072	830	1.40	UC Nazimabad	Liaquatabad	Karachi Central
NULL	NULL	NULL	BLK-UC071	810	1.60	UC Liaquatabad Central	Liaquatabad	Karachi Central
NULL	NULL	NULL	BLK-UC061	800	4.50	UC Clifton	Saddar (KHI South)	Karachi South
NULL	NULL	NULL	BLK-UC063	750	1.80	UC Saddar KHI	Aram Bagh	Karachi South
NULL	NULL	NULL	BLK-UC193	750	2.00	UC F-7	F-Series Sectors	Islamabad Urban
NULL	NULL	NULL	BLK-UC194	720	2.20	UC F-8	F-Series Sectors	Islamabad Urban
NULL	NULL	NULL	BLK-UC066	710	2.80	UC Pehlwan Goth	Gulshan-e-Iqbal	Karachi East
NULL	NULL	NULL	BLK-UC065	680	3.20	UC Gulshan Central	Gulshan-e-Iqbal	Karachi East
NULL	NULL	NULL	BLK-UC195	680	2.50	UC G-9	G-Series Sectors	Islamabad Urban
NULL	NULL	NULL	BLK-UC196	650	2.80	UC G-10	G-Series Sectors	Islamabad Urban
NULL	NULL	NULL	BLK-UC068	640	2.40	UC Tariq Road	Ferozabad	Karachi East
NULL	NULL	NULL	BLK-UC067	620	2.50	UC PECHS	Ferozabad	Karachi East
NULL	NULL	NULL	BLK-UC157	620	2.20	UC Nawa Killi	Quetta City	Quetta
NULL	NULL	NULL	BLK-UC117	600	6.50	UC Nowshera Cantt	Nowshera City	Nowshera
NULL	NULL	NULL	BLK-UC158	600	2.40	UC Satellite Town QTA	Quetta City	Quetta
NULL	NULL	NULL	BLK-UC109	580	3.50	UC Hayatabad	Peshawar City	Peshawar
NULL	NULL	NULL	BLK-UC118	580	7.00	UC Risalpur	Nowshera City	Nowshera
NULL	NULL	NULL	BLK-UC152	580	5.50	UC Tarbela Dam Sector	Topi	Swabi
NULL	NULL	NULL	BLK-UC145	570	2.50	UC Mardan Central	Mardan City	Mardan
NULL	NULL	NULL	BLK-UC110	560	3.80	UC University Town	Peshawar City	Peshawar
Bushra Ansari	Enumerator	0300-1000035	BLK-UC032	550	6.20	UC Shorkot Cantt	Shorkot	Jhang
NULL	NULL	NULL	BLK-UC082	550	5.20	UC SITE Kotri	Kotri	Jamshoro
NULL	NULL	NULL	BLK-UC146	550	2.70	UC Hoti	Mardan City	Mardan
NULL	NULL	NULL	BLK-UC137	540	2.60	UC Mansehra Central	Mansehra City	Mansehra
NULL	NULL	NULL	BLK-UC097	530	2.40	UC Larkana Central	Larkana City	Larkana
NULL	NULL	NULL	BLK-UC073	520	3.50	UC Latifabad	Hyderabad City	Hyderabad
NULL	NULL	NULL	BLK-UC144	520	6.00	UC Tarbela	Ghazi	Haripur
NULL	NULL	NULL	BLK-UC098	510	2.60	UC Dari	Larkana City	Larkana
NULL	NULL	NULL	BLK-UC121	510	2.80	UC Mingora City	Babuzai (Mingora)	Swat

Shows educated and full-time employed persons with their religion and mother tongue profile.

```
mysql> SELECT
    ->     CONCAT(p.first_name, ' ', p.last_name) AS name,
    ->     r.religion_name,
    ->     mt.tongue_name AS mother_tongue,
    ->     el.level_name AS education_level,
    ->     es.status_name AS employment_status,
    ->     p.monthly_income,
    ->     TIMESTAMPDIFF(YEAR, p.date_of_birth, CURDATE()) AS age
    -> FROM Religion r
    -> RIGHT JOIN Person p ON r.religion_id = p.religion_id
    -> RIGHT JOIN Household h ON p.household_id = h.household_id
    -> RIGHT JOIN Mother_Tongue mt ON p.mother_tongue_id = mt.mother_tongue_id
    -> RIGHT JOIN Education_Level el ON p.education_level_id = el.education_level_id
    -> LEFT JOIN Employment_Status es ON p.employment_status_id = es.employment_status_id
    -> WHERE UPPER(el.level_name) IN ('INTERMEDIATE', 'BACHELORS', 'MASTERS', 'PHD') AND es.status_name = 'Employed Full-Time' AND p.monthly_income > 30000
    -> ORDER BY p.monthly_income DESC;
```

name	religion_name	mother_tongue	education_level	employment_status	monthly_income	age
Khalid Ahmed	Islam	Urdu	Bachelors	Employed Full-Time	55000.00	40
Muhammad Khan	Islam	Urdu	Masters	Employed Full-Time	45000.00	61
Malik Hassan	Islam	Punjabi	PHD	Employed Full-Time	35000.00	33
Hamza Hassan	Christianity	Hindko	Intermediate	Employed Full-Time	34000.00	34
Kamil Ahmad	Islam	Urdu	Intermediate	Employed Full-Time	33000.00	30
Mir Khan	Islam	Saraiki	Intermediate	Employed Full-Time	32000.00	32
Karim Ali	Islam	Urdu	Intermediate	Employed Full-Time	32000.00	32
Farhan Hassan	Islam	Urdu	Intermediate	Employed Full-Time	31000.00	32

8 rows in set (0.01 sec)

NATURAL JOIN

Shows married individuals with an income above 20,000, joined implicitly on marital_status_id.

```
mysql> SELECT
    ->     p.first_name,
    ->     p.last_name,
    ->     p.date_of_birth,
    ->     p.monthly_income,
    ->     ms.status_name
    -> FROM Marital_Status ms
    -> NATURAL JOIN (
    ->     SELECT marital_status_id, first_name, last_name, date_of_birth, monthly_income
    ->     FROM Person
    -> ) p
    -> WHERE ms.status_name = 'Married' AND p.monthly_income > 20000
    -> ORDER BY p.monthly_income DESC;
+------------+-----------+---------------+----------------+-------------+
| first_name | last_name | date_of_birth | monthly_income | status_name |
+------------+-----------+---------------+----------------+-------------+
| Yousef     | Ali       | 1958-01-08    |       72000.00 | Married     |
| Rashad     | Ahmed     | 1959-09-19    |       71000.00 | Married     |
| Faisal     | Hassan    | 1956-12-17    |       68000.00 | Married     |
| Waqar      | Ahmed     | 1955-12-10    |       65000.00 | Married     |
| Zahir      | Ahmad     | 1960-06-14    |       62000.00 | Married     |
| Shabir     | Khan      | 1958-01-25    |       60000.00 | Married     |
| Mohan      | Hassan    | 1957-05-12    |       58000.00 | Married     |
| Riaz       | Ali       | 1958-07-15    |       55000.00 | Married     |
| Nadim      | Khan      | 1968-09-02    |       53000.00 | Married     |
| Samir      | Khan      | 1970-04-20    |       52000.00 | Married     |
| Zubair     | Hassan    | 1966-03-14    |       51000.00 | Married     |
| Wasim      | Malik     | 1969-07-09    |       51000.00 | Married     |
| Akhtar     | Hassan    | 1963-04-08    |       50000.00 | Married     |
| Habib      | Malik     | 1967-03-06    |       49000.00 | Married     |
| Nasir      | Hussain   | 1962-11-05    |       48000.00 | Married     |
| Manzoor    | Ahmad     | 1969-05-21    |       48000.00 | Married     |
| Salim      | Ahmed     | 1965-02-09    |       47000.00 | Married     |
| Nawaz      | Ali       | 1964-07-06    |       46000.00 | Married     |
| Adnan      | Ahmed     | 1961-10-25    |       46000.00 | Married     |
| Muhammad   | Khan      | 1965-03-15    |       45000.00 | Married     |
| Deepak     | Ali       | 1964-02-22    |       44000.00 | Married     |
| Feroz      | Malik     | 1961-11-20    |       44000.00 | Married     |
| Babar      | Khan      | 1962-11-28    |       43000.00 | Married     |
| Rashid     | Hassan    | 1960-01-22    |       42000.00 | Married     |
| Jalal      | Khan      | 1962-08-10    |       42000.00 | Married     |
| Jamal      | Malik     | 1972-06-18    |       41000.00 | Married     |
| Ibrahim    | Raza      | 1959-08-03    |       40000.00 | Married     |
| Qasim      | Malik     | 1960-10-11    |       39000.00 | Married     |
| Tariq      | Hassan    | 1963-08-29    |       39000.00 | Married     |
| Ahmed      | Ali       | 1968-07-19    |       38000.00 | Married     |
| Saad       | Khan      | 1966-04-17    |       37000.00 | Married     |
| Siddiqui   | Khan      | 1961-09-30    |       35000.00 | Married     |
+------------+-----------+---------------+----------------+-------------+
32 rows in set (0.00 sec)
```

Shows individuals with premium employment statuses earning over 30,000, joined implicitly on employment_status_id.

```
mysql> SELECT
    ->     p.first_name,
    ->     p.last_name,
    ->     p.monthly_income,
    ->     p.date_of_birth,
    ->     es.status_name
    -> FROM Employment_Status es
    -> NATURAL JOIN (
    ->     SELECT employment_status_id, first_name, last_name, monthly_income, date_of_birth
    ->     FROM Person
    -> ) p
    -> WHERE es.status_name IN ('Employed Full-Time', 'Self-Employed') AND p.monthly_income > 30000
    -> ORDER BY p.monthly_income DESC;
+------------+-----------+----------------+---------------+--------------------+
| first_name | last_name | monthly_income | date_of_birth | status_name        |
+------------+-----------+----------------+---------------+--------------------+
| Rashad     | Ahmed     |       71000.00 | 1959-09-19    | Employed Full-Time |
| Faisal     | Hassan    |       68000.00 | 1956-12-17    | Employed Full-Time |
| Zahir      | Ahmad     |       62000.00 | 1960-06-14    | Employed Full-Time |
| Shabir     | Khan      |       60000.00 | 1958-01-25    | Employed Full-Time |
| Mohan      | Hassan    |       58000.00 | 1957-05-12    | Employed Full-Time |
| Riaz       | Ali       |       55000.00 | 1958-07-15    | Employed Full-Time |
| Khalid     | Ahmed     |       55000.00 | 1985-09-13    | Employed Full-Time |
| Nadim      | Khan      |       53000.00 | 1968-09-02    | Employed Full-Time |
| Samir      | Khan      |       52000.00 | 1970-04-20    | Employed Full-Time |
| Zubair     | Hassan    |       51000.00 | 1966-03-14    | Employed Full-Time |
| Wasim      | Malik     |       51000.00 | 1969-07-09    | Employed Full-Time |
| Akhtar     | Hassan    |       50000.00 | 1963-04-08    | Employed Full-Time |
| Habib      | Malik     |       49000.00 | 1967-03-06    | Employed Full-Time |
| Nasir      | Hussain   |       48000.00 | 1962-11-05    | Employed Full-Time |
| Manzoor    | Ahmad     |       48000.00 | 1969-05-21    | Employed Full-Time |
| Salim      | Ahmed     |       47000.00 | 1965-02-09    | Employed Full-Time |
| Nawaz      | Ali       |       46000.00 | 1964-07-06    | Employed Full-Time |
| Adnan      | Ahmed     |       46000.00 | 1961-10-25    | Employed Full-Time |
| Muhammad   | Khan      |       45000.00 | 1965-03-15    | Employed Full-Time |
| Feroz      | Malik     |       44000.00 | 1961-11-20    | Employed Full-Time |
| Deepak     | Ali       |       44000.00 | 1964-02-22    | Employed Full-Time |
| Babar      | Khan      |       43000.00 | 1962-11-28    | Employed Full-Time |
| Rashid     | Hassan    |       42000.00 | 1960-01-22    | Employed Full-Time |
| Jalal      | Khan      |       42000.00 | 1962-08-10    | Employed Full-Time |
| Jamal      | Malik     |       41000.00 | 1972-06-18    | Employed Full-Time |
| Ibrahim    | Raza      |       40000.00 | 1959-08-03    | Employed Full-Time |
| Tariq      | Hassan    |       39000.00 | 1963-08-29    | Employed Full-Time |
| Qasim      | Malik     |       39000.00 | 1960-10-11    | Employed Full-Time |
| Ahmed      | Ali       |       38000.00 | 1968-07-19    | Employed Full-Time |
| Saad       | Khan      |       37000.00 | 1966-04-17    | Employed Full-Time |
| Ashan      | Ahmed     |       36000.00 | 1993-06-22    | Employed Full-Time |
| Siddiqui   | Khan      |       35000.00 | 1961-09-30    | Employed Full-Time |
| Malik      | Hassan    |       35000.00 | 1992-10-30    | Employed Full-Time |
| Hamza      | Hassan    |       34000.00 | 1991-10-22    | Employed Full-Time |
| Kamil      | Ahmad     |       33000.00 | 1996-02-10    | Employed Full-Time |
| Mir        | Khan      |       32000.00 | 1994-03-29    | Employed Full-Time |
| Karim      | Ali       |       32000.00 | 1993-12-05    | Employed Full-Time |
```

SELF JOIN

Maps out the internal management hierarchy of the census field staff team.

```
mysql> SELECT
    ->     CONCAT(emp.first_name, ' ', emp.last_name) AS staff_member,
    ->     emp.staff_role AS Staff_Role,
    ->     CONCAT(mgr.first_name, ' ', mgr.last_name) AS direct_supervisor
    -> FROM Census_Staff emp
    -> INNER JOIN Census_Staff mgr ON emp.supervisor_id = mgr.staff_id;
+----------------+---------------------+-------------------+
| staff_member   | Staff_Role          | direct_supervisor |
+----------------+---------------------+-------------------+
| Ali Hassan     | Enumerator          | Ahmed Raza        |
| Zainab Noor    | Enumerator          | Ahmed Raza        |
| Bilal Khan     | Enumerator          | Fatima Bibi       |
| Maria Siddiqui | Enumerator          | Fatima Bibi       |
| Omar Farooq    | Enumerator          | Tariq Mehmood     |
| Sana Javed     | Enumerator          | Tariq Mehmood     |
| Usman Ghani    | Enumerator          | Ahmed Raza        |
| Hina Naz       | Enumerator          | Ahmed Raza        |
| Imran Shah     | Enumerator          | Fatima Bibi       |
| Sadaf Ali      | Enumerator          | Fatima Bibi       |
| Hamza Rehman   | Enumerator          | Tariq Mehmood     |
| Aqsa Iqbal     | Enumerator          | Tariq Mehmood     |
| Kashif Latif   | Data_Entry_Operator | Ahmed Raza        |
| Rabia Basri    | Data_Entry_Operator | Ahmed Raza        |
| Waqas Ahmed    | Data_Entry_Operator | Fatima Bibi       |
| Nadia Gul      | Data_Entry_Operator | Fatima Bibi       |
| Zeeshan Malik  | Data_Entry_Operator | Tariq Mehmood     |
| Amna Parveen   | Data_Entry_Operator | Tariq Mehmood     |
| Salman Farsi   | Quality_Checker     | Ahmed Raza        |
| Sadia Imran    | Quality_Checker     | Ahmed Raza        |
| Faizan Aslam   | Quality_Checker     | Fatima Bibi       |
| Sidra Munir    | Quality_Checker     | Fatima Bibi       |
| Adnan Sami     | Quality_Checker     | Tariq Mehmood     |
| Madiha Khalid  | Quality_Checker     | Tariq Mehmood     |
| Shahid Afidi   | Enumerator          | Ahmed Raza        |
| Mehwish Hayat  | Enumerator          | Ahmed Raza        |
| Daniyal Arshad | Enumerator          | Fatima Bibi       |
| Arooj Fatima   | Enumerator          | Fatima Bibi       |
| Bilal Saeed    | Enumerator          | Tariq Mehmood     |
| Anum Zehra     | Enumerator          | Tariq Mehmood     |
| Naveed Ul-Haq  | Enumerator          | Ahmed Raza        |
| Bushra Ansari  | Enumerator          | Ahmed Raza        |
| Junaid Jamshed | Enumerator          | Fatima Bibi       |
| Sumaira Khan   | Enumerator          | Fatima Bibi       |
| Fahad Mustafa  | Enumerator          | Tariq Mehmood     |
| Maryam Nawaz   | Enumerator          | Tariq Mehmood     |
| Saqib Mehmood  | Enumerator          | Ahmed Raza        |
| Saba Qamar     | Enumerator          | Ahmed Raza        |
| Waseem Akram   | Enumerator          | Fatima Bibi       |
| Sofia Khan     | Enumerator          | Fatima Bibi       |
| Yasir Shah     | Enumerator          | Tariq Mehmood     |
| Ambar Bashir   | Enumerator          | Tariq Mehmood     |
| Zahid Iqbal    | Enumerator          | Ahmed Raza        |
| Shazia Parveen | Enumerator          | Ahmed Raza        |
```

Maps out biological family dynamics by identifying the official household head for each member.

```
mysql> SELECT
    ->     CONCAT(member.first_name, ' ', member.last_name) AS family_member,
    ->     member.cnic_number AS Member_CNIC,
    ->     CONCAT(head.first_name, ' ', head.last_name) AS household_head,
    ->     head.cnic_number AS Head_CNIC
    -> FROM Person member
    -> INNER JOIN Person head ON member.head_person_id = head.person_id
    -> WHERE member.person_id <> member.head_person_id;
+----------------+-----------------+----------------+-----------------+
| family_member  | Member_CNIC     | household_head | Head_CNIC       |
+----------------+-----------------+----------------+-----------------+
| Fatima Khan    | 12345-6000002-3 | Muhammad Khan  | 12345-6000001-1 |
| Ali Khan       | 12345-6000003-5 | Muhammad Khan  | 12345-6000001-1 |
| Ayesha Khan    | 12345-6000004-7 | Muhammad Khan  | 12345-6000001-1 |
| Hassan Khan    | 12345-6000005-9 | Muhammad Khan  | 12345-6000001-1 |
| Noor Ali       | 12345-6000007-2 | Ahmed Ali      | 12345-6000006-0 |
| Karim Ali      | 12345-6000008-4 | Ahmed Ali      | 12345-6000006-0 |
| Sara Ali       | 12345-6000009-6 | Ahmed Ali      | 12345-6000006-0 |
| Zainab Hassan  | 12345-6000011-0 | Rashid Hassan  | 12345-6000010-8 |
| Malik Hassan   | 12345-6000012-2 | Rashid Hassan  | 12345-6000010-8 |
| Hina Hassan    | 12345-6000013-4 | Rashid Hassan  | 12345-6000010-8 |
| Amina Hussain  | 12345-6000015-8 | Nasir Hussain  | 12345-6000014-6 |
| Imran Hussain  | 12345-6000016-1 | Nasir Hussain  | 12345-6000014-6 |
| Leila Hussain  | 12345-6000017-3 | Nasir Hussain  | 12345-6000014-6 |
| Maha Khan      | 12345-6000019-7 | Samir Khan     | 12345-6000018-5 |
| Bashir Khan    | 12345-6000020-9 | Samir Khan     | 12345-6000018-5 |
| Rania Ahmed    | 12345-6000022-2 | Waqar Ahmed    | 12345-6000021-0 |
| Khalid Ahmed   | 12345-6000023-4 | Waqar Ahmed    | 12345-6000021-0 |
| Dina Ahmed     | 12345-6000024-6 | Waqar Ahmed    | 12345-6000021-0 |
| Lia Malik      | 12345-6000026-1 | Jamal Malik    | 12345-6000025-8 |
| Amir Malik     | 12345-6000027-3 | Jamal Malik    | 12345-6000025-8 |
| Nadia Hassan   | 12345-6000029-7 | Tariq Hassan   | 12345-6000028-5 |
| Farhan Hassan  | 12345-6000030-9 | Tariq Hassan   | 12345-6000028-5 |
| Hana Ali       | 12345-6000032-2 | Yousef Ali     | 12345-6000031-0 |
| Rasheed Ali    | 12345-6000033-4 | Yousef Ali     | 12345-6000031-0 |
| Luna Khan      | 12345-6000035-8 | Saad Khan      | 12345-6000034-6 |
| Aziz Khan      | 12345-6000036-1 | Saad Khan      | 12345-6000034-6 |
| Yasmin Ahmed   | 12345-6000038-5 | Adnan Ahmed    | 12345-6000037-3 |
| Rana Ahmed     | 12345-6000039-7 | Adnan Ahmed    | 12345-6000037-3 |
| Rena Malik     | 12345-6000041-0 | Wasim Malik    | 12345-6000040-9 |
| Naeem Malik    | 12345-6000042-2 | Wasim Malik    | 12345-6000040-9 |
| Anjali Hassan  | 12345-6000044-6 | Mohan Hassan   | 12345-6000043-4 |
| Rohan Hassan   | 12345-6000045-8 | Mohan Hassan   | 12345-6000043-4 |
| Priya Ali      | 12345-6000047-3 | Deepak Ali     | 12345-6000046-1 |
| Arjun Ali      | 12345-6000048-5 | Deepak Ali     | 12345-6000046-1 |
| Gulnar Khan    | 12345-6000050-9 | Siddiqui Khan  | 12345-6000049-7 |
| Farah Khan     | 12345-6000051-0 | Siddiqui Khan  | 12345-6000049-7 |
| Meena Ahmad    | 12345-6000053-4 | Zahir Ahmad    | 12345-6000052-2 |
| Kamil Ahmad    | 12345-6000054-6 | Zahir Ahmad    | 12345-6000052-2 |
| Saira Raza     | 12345-6000056-1 | Ibrahim Raza   | 12345-6000055-8 |
| Kashan Raza    | 12345-6000057-3 | Ibrahim Raza   | 12345-6000055-8 |
| Mariam Khan    | 12345-6000059-7 | Babar Khan     | 12345-6000058-5 |
```

B. Nested and Correlated Queries

Retrieves names of citizens who live in crowded families with more than 4 members.

```
mysql> SELECT first_name, last_name
    -> FROM Person
    -> WHERE household_id IN (
    ->     SELECT household_id
    ->     FROM Household
    ->     WHERE household_size > 4
    -> );
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Muhammad   | Khan      |
| Fatima     | Khan      |
| Ali        | Khan      |
| Ayesha     | Khan      |
| Hassan     | Khan      |
+------------+-----------+
5 rows in set (0.00 sec)
```

Finds citizens earning more than at least one individual of biological sex male.

```
mysql> SELECT first_name, monthly_income
    -> FROM Person
    -> WHERE monthly_income > ANY (
    ->     SELECT monthly_income
    ->     FROM Person
    ->     WHERE sex_id = 2
    -> );
+------------+----------------+
| first_name | monthly_income |
+------------+----------------+
| Muhammad   |       45000.00 |
| Ahmed      |       38000.00 |
| Karim      |       32000.00 |
| Rashid     |       42000.00 |
| Malik      |       35000.00 |
| Nasir      |       48000.00 |
| Imran      |       28000.00 |
| Samir      |       52000.00 |
| Waqar      |       65000.00 |
| Khalid     |       55000.00 |
| Dina       |       25000.00 |
| Jamal      |       41000.00 |
| Tariq      |       39000.00 |
| Farhan     |       31000.00 |
| Yousef     |       72000.00 |
| Rasheed    |       29000.00 |
| Saad       |       37000.00 |
| Adnan      |       46000.00 |
| Rana       |       24000.00 |
| Wasim      |       51000.00 |
| Mohan      |       58000.00 |
| Rohan      |       26000.00 |
| Deepak     |       44000.00 |
| Siddiqui   |       35000.00 |
| Farah      |       18000.00 |
| Zahir      |       62000.00 |
| Kamil      |       33000.00 |
| Ibrahim    |       40000.00 |
| Kashan     |       23000.00 |
| Babar      |       43000.00 |
| Danial     |       27000.00 |
| Habib      |       49000.00 |
| Faisal     |       68000.00 |
```

Finds citizens earning more than all females

```
mysql> SELECT first_name, monthly_income
    -> FROM Person
    -> WHERE monthly_income > ALL (
    ->     SELECT monthly_income
    ->     FROM Person
    ->     WHERE sex_id = 2 AND monthly_income IS NOT NULL
    -> );
+------------+----------------+
| first_name | monthly_income |
+------------+----------------+
| Muhammad   |       45000.00 |
| Ahmed      |       38000.00 |
| Karim      |       32000.00 |
| Rashid     |       42000.00 |
| Malik      |       35000.00 |
| Nasir      |       48000.00 |
| Imran      |       28000.00 |
| Samir      |       52000.00 |
| Waqar      |       65000.00 |
| Khalid     |       55000.00 |
| Jamal      |       41000.00 |
| Tariq      |       39000.00 |
| Farhan     |       31000.00 |
| Yousef     |       72000.00 |
| Rasheed    |       29000.00 |
| Saad       |       37000.00 |
| Adnan      |       46000.00 |
| Wasim      |       51000.00 |
| Mohan      |       58000.00 |
| Rohan      |       26000.00 |
| Deepak     |       44000.00 |
| Siddiqui   |       35000.00 |
| Zahir      |       62000.00 |
| Kamil      |       33000.00 |
| Ibrahim    |       40000.00 |
| Babar      |       43000.00 |
| Danial     |       27000.00 |
| Habib      |       49000.00 |
| Faisal     |       68000.00 |
| Hamza      |       34000.00 |
| Salim      |       47000.00 |
| Riaz       |       55000.00 |
| Nadim      |       53000.00 |
| Qasim      |       39000.00 |
```

Lists all citizens who have never migrated or shifted their residential district.

```
mysql> SELECT first_name, last_name
    -> FROM Person
    -> WHERE person_id NOT IN (
    ->     SELECT person_id
    ->     FROM Migration_History
    -> );
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Farah      | Khan      |
| Zahir      | Ahmad     |
| Meena      | Ahmad     |
| Kamil      | Ahmad     |
| Ibrahim    | Raza      |
| Saira      | Raza      |
| Kashan     | Raza      |
| Babar      | Khan      |
| Mariam     | Khan      |
| Danial     | Khan      |
| Habib      | Malik     |
| Zara       | Malik     |
| Karim      | Malik     |
| Faisal     | Hassan    |
| Iqra       | Hassan    |
| Hamza      | Hassan    |
| Salim      | Ahmed     |
| Saba       | Ahmed     |
| Hassam     | Ahmed     |
| Riaz       | Ali       |
| Bushra     | Ali       |
| Imad       | Ali       |
| Nadim      | Khan      |
| Suhana     | Khan      |
| Shiraz     | Khan      |
| Qasim      | Malik     |
| Hafsa      | Malik     |
```

Finds citizens earning less than or equal to the absolute lowest salary among males.

```
mysql> SELECT first_name, last_name
    -> FROM Person
    -> WHERE monthly_income <= ALL (
    ->     SELECT monthly_income
    ->     FROM Person
    ->     WHERE sex_id = 1
    -> );
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Ali        | Khan      |
| Ayesha     | Khan      |
| Hassan     | Khan      |
| Sara       | Ali       |
| Hina       | Hassan    |
| Leila      | Hussain   |
| Bashir     | Khan      |
| Amir       | Malik     |
| Aziz       | Khan      |
| Naeem      | Malik     |
| Arjun      | Ali       |
| Karim      | Malik     |
| Hassam     | Ahmed     |
| Imad       | Ali       |
| Shiraz     | Khan      |
| Sarmad     | Ahmad     |
| Rizwan     | Ali       |
| Suleman    | Hassan    |
| Shafiq     | Khan      |
+------------+-----------+
19 rows in set (0.00 sec)
```

Finds citizens earning less than or equal to at least one individual of among males.

```
mysql> SELECT first_name, last_name
    -> FROM Person
    -> WHERE monthly_income <= ANY (
    ->     SELECT monthly_income
    ->     FROM Person
    ->     WHERE sex_id = 1
    -> );
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Muhammad   | Khan      |
| Ali        | Khan      |
| Ayesha     | Khan      |
| Hassan     | Khan      |
| Ahmed      | Ali       |
| Karim      | Ali       |
| Sara       | Ali       |
| Rashid     | Hassan    |
| Malik      | Hassan    |
| Hina       | Hassan    |
| Nasir      | Hussain   |
| Imran      | Hussain   |
| Leila      | Hussain   |
| Samir      | Khan      |
| Bashir     | Khan      |
| Waqar      | Ahmed     |
| Khalid     | Ahmed     |
| Dina       | Ahmed     |
| Jamal      | Malik     |
| Amir       | Malik     |
| Tariq      | Hassan    |
| Farhan     | Hassan    |
| Yousef     | Ali       |
| Rasheed    | Ali       |
| Saad       | Khan      |
| Aziz       | Khan      |
| Adnan      | Ahmed     |
```

Identifies citizens who have an active internal migration or relocation history record.

```
mysql> SELECT p.first_name, p.last_name
    -> FROM Person p
    -> WHERE EXISTS (
    ->     SELECT 1
    ->     FROM Migration_History m
    ->     WHERE m.person_id = p.person_id
    -> );
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Muhammad   | Khan      |
| Fatima     | Khan      |
| Ali        | Khan      |
| Ayesha     | Khan      |
| Hassan     | Khan      |
| Ahmed      | Ali       |
| Noor       | Ali       |
| Karim      | Ali       |
| Sara       | Ali       |
| Rashid     | Hassan    |
| Zainab     | Hassan    |
| Malik      | Hassan    |
| Hina       | Hassan    |
| Nasir      | Hussain   |
| Amina      | Hussain   |
| Imran      | Hussain   |
| Leila      | Hussain   |
| Samir      | Khan      |
| Maha       | Khan      |
| Bashir     | Khan      |
| Waqar      | Ahmed     |
| Rania      | Ahmed     |
| Khalid     | Ahmed     |
| Dina       | Ahmed     |
| Jamal      | Malik     |
| Lia        | Malik     |
| Amir       | Malik     |
| Tariq      | Hassan    |
| Nadia      | Hassan    |
```

Identifies stable citizens who have zero documented history of changing districts.

```
mysql> SELECT p.first_name, p.last_name
    -> FROM Person p
    -> WHERE NOT EXISTS (
    ->     SELECT 1
    ->     FROM Migration_History m
    ->     WHERE m.person_id = p.person_id
    -> );
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Farah      | Khan      |
| Zahir      | Ahmad     |
| Meena      | Ahmad     |
| Kamil      | Ahmad     |
| Ibrahim    | Raza      |
| Saira      | Raza      |
| Kashan     | Raza      |
| Babar      | Khan      |
| Mariam     | Khan      |
| Danial     | Khan      |
| Habib      | Malik     |
| Zara       | Malik     |
| Karim      | Malik     |
| Faisal     | Hassan    |
| Iqra       | Hassan    |
| Hamza      | Hassan    |
| Salim      | Ahmed     |
| Saba       | Ahmed     |
| Hassam     | Ahmed     |
| Riaz       | Ali       |
| Bushra     | Ali       |
| Imad       | Ali       |
| Nadim      | Khan      |
| Suhana     | Khan      |
| Shiraz     | Khan      |
| Qasim      | Malik     |
| Hafsa      | Malik     |
```

C. Aggregate Queries

Calculates the total headcount of recorded citizens in the census system database.

```
mysql> SELECT COUNT(*) AS TotalPopulation
    -> FROM Person;
+-----------------+
| TotalPopulation |
+-----------------+
|             102 |
+-----------------+
1 row in set (0.01 sec)
```

Computes the gross combined monthly income of the entire surveyed population.

```
mysql> SELECT SUM(monthly_income) AS TotalIncome
    -> FROM Person;
+-------------+
| TotalIncome |
+-------------+
|  2149000.00 |
+-------------+
1 row in set (0.00 sec)
```

Evaluates the national baseline average monthly income across all citizens.

```
mysql> SELECT AVG(monthly_income) AS AverageIncome
    -> FROM Person;
+---------------+
| AverageIncome |
+---------------+
|  30700.000000 |
+---------------+
1 row in set (0.00 sec)
```

Pinpoints the absolute lowest recorded individual monthly income entry.

```
mysql>
mysql> SELECT MIN(monthly_income) AS LowestIncome
    -> FROM Person;
+--------------+
| LowestIncome |
+--------------+
|         0.00 |
+--------------+
1 row in set (0.00 sec)
```

Pinpoints the absolute highest recorded individual monthly income entry.

```
mysql> SELECT MAX(monthly_income) AS HighestIncome
    -> FROM Person;
+---------------+
| HighestIncome |
+---------------+
|      72000.00 |
+---------------+
1 row in set (0.00 sec)
```

Groups and counts total population distribution cleanly across distinct biological sexes.

```
mysql> SELECT b.sex_name, COUNT(*) AS Population
    -> FROM Person p
    -> INNER JOIN Biological_Sex b ON p.sex_id = b.sex_id
    -> GROUP BY b.sex_name;
+----------+------------+
| sex_name | Population |
+----------+------------+
| Male     |         64 |
| Female   |         38 |
+----------+------------+
2 rows in set (0.00 sec)
```

Filters demographic categories to show only biological sexes with more than 40 individuals.

```
mysql> SELECT sex_id, COUNT(*) AS Population
    -> FROM Person
    -> GROUP BY sex_id
    -> HAVING COUNT(*) > 40;
+--------+------------+
| sex_id | Population |
+--------+------------+
|      1 |         64 |
+--------+------------+
1 row in set (0.00 sec)
```

Breaks down population counts and average monthly income metrics categorized by religion.

```
mysql> SELECT r.religion_name, COUNT(*) AS Population, AVG(monthly_income) AS AvgIncome
    -> FROM Person p
    -> INNER JOIN Religion r ON p.religion_id = r.religion_id
    -> GROUP BY r.religion_name;
+---------------+------------+--------------+
| religion_name | Population | AvgIncome    |
+---------------+------------+--------------+
| Islam         |         93 | 29984.375000 |
| Hinduism      |          6 | 32000.000000 |
| Christianity  |          3 | 51000.000000 |
+---------------+------------+--------------+
3 rows in set (0.01 sec)
```

Combines distinct first names from both citizens and field staff, removing duplicate values.

```
mysql> SELECT first_name, last_name FROM Person
    -> UNION
    -> SELECT first_name, last_name FROM Census_Staff;
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Muhammad   | Khan      |
| Fatima     | Khan      |
| Ali        | Khan      |
| Ayesha     | Khan      |
| Hassan     | Khan      |
| Ahmed      | Ali       |
| Noor       | Ali       |
| Karim      | Ali       |
| Sara       | Ali       |
| Rashid     | Hassan    |
| Zainab     | Hassan    |
| Malik      | Hassan    |
| Hina       | Hassan    |
| Nasir      | Hussain   |
| Amina      | Hussain   |
| Imran      | Hussain   |
| Leila      | Hussain   |
| Samir      | Khan      |
| Maha       | Khan      |
| Bashir     | Khan      |
| Waqar      | Ahmed     |
| Rania      | Ahmed     |
| Khalid     | Ahmed     |
| Dina       | Ahmed     |
| Jamal      | Malik     |
| Lia        | Malik     |
| Amir       | Malik     |
| Tariq      | Hassan    |
| Nadia      | Hassan    |
| Farhan     | Hassan    |
```

Combines all first names from citizens and field staff, preserving every duplicate entry.

```
mysql> SELECT first_name, last_name FROM Person
    -> UNION ALL
    -> SELECT first_name, last_name FROM Census_Staff;
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Muhammad   | Khan      |
| Fatima     | Khan      |
| Ali        | Khan      |
| Ayesha     | Khan      |
| Hassan     | Khan      |
| Ahmed      | Ali       |
| Noor       | Ali       |
| Karim      | Ali       |
| Sara       | Ali       |
| Rashid     | Hassan    |
| Zainab     | Hassan    |
| Malik      | Hassan    |
| Hina       | Hassan    |
| Nasir      | Hussain   |
| Amina      | Hussain   |
| Imran      | Hussain   |
| Leila      | Hussain   |
| Samir      | Khan      |
| Maha       | Khan      |
| Bashir     | Khan      |
| Waqar      | Ahmed     |
| Rania      | Ahmed     |
| Khalid     | Ahmed     |
| Dina       | Ahmed     |
| Jamal      | Malik     |
| Lia        | Malik     |
| Amir       | Malik     |
| Tariq      | Hassan    |
| Nadia      | Hassan    |
| Farhan     | Hassan    |
| Yousef     | Ali       |
| Hana       | Ali       |
```

Finds common names shared by both active citizens and field census staff.

```
mysql> SELECT first_name, last_name
    -> FROM Person
    -> WHERE first_name IN (
    ->     SELECT first_name
    ->     FROM Census_Staff
    -> );
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Fatima     | Khan      |
| Ali        | Khan      |
| Ahmed      | Ali       |
| Zainab     | Hassan    |
| Hina       | Hassan    |
| Imran      | Hussain   |
| Tariq      | Hassan    |
| Nadia      | Hassan    |
| Adnan      | Ahmed     |
| Hamza      | Hassan    |
| Saba       | Ahmed     |
| Bushra     | Ali       |
| Rabia      | Hassan    |
| Rizwan     | Ali       |
+------------+-----------+
14 rows in set (0.01 sec)
```

Lists first names unique to citizens that do not exist among census staff.

```
mysql> SELECT first_name, last_name
    -> FROM Person
    -> WHERE first_name NOT IN (
    ->     SELECT first_name
    ->     FROM Census_Staff
    -> );
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Muhammad   | Khan      |
| Ayesha     | Khan      |
| Hassan     | Khan      |
| Noor       | Ali       |
| Karim      | Ali       |
| Sara       | Ali       |
| Rashid     | Hassan    |
| Malik      | Hassan    |
| Nasir      | Hussain   |
| Amina      | Hussain   |
| Leila      | Hussain   |
| Samir      | Khan      |
| Maha       | Khan      |
| Bashir     | Khan      |
| Waqar      | Ahmed     |
| Rania      | Ahmed     |
| Khalid     | Ahmed     |
| Dina       | Ahmed     |
| Jamal      | Malik     |
| Lia        | Malik     |
| Amir       | Malik     |
| Farhan     | Hassan    |
| Yousef     | Ali       |
| Hana       | Ali       |
| Rasheed    | Ali       |
| Saad       | Khan      |
| Luna       | Khan      |
| Aziz       | Khan      |
| Yasmin     | Ahmed     |
```

D. Views

Shows a complete profile of each citizen, including their full name, age, education, job, and income in plain words instead of codes.

```
mysql> SELECT * FROM Citizen_SocioEconomic_Profile;
```

person_id	full_name	cnic_number	biological_sex	age	religion	education_level	employment_status	occupation	industry	monthly_income
1	Muhammad Khan	12345-6000001-1	Male	61	Islam	Masters	Employed Full-Time	Doctor/Medical Professional	Public Administration	45000.00
2	Fatima Khan	12345-6000002-3	Female	56	Islam	PHD	Homemaker	NULL	NULL	NULL
3	Ali Khan	12345-6000003-5	Male	30	Islam	Masters	Student	Government Official	Wholesale and Retail Trade	0.00
4	Ayesha Khan	12345-6000004-7	Female	27	Islam	Intermediate	Student	NULL	NULL	0.00
5	Hassan Khan	12345-6000005-9	Male	24	Islam	Primary	Student	NULL	NULL	0.00
6	Ahmed Ali	12345-6000006-0	Male	57	Islam	Matriculation	Employed Full-Time	Driver/Transport Worker	Agriculture and Farming	38000.00
7	Noor Ali	12345-6000007-2	Female	53	Islam	Middle	Homemaker	NULL	NULL	NULL
8	Karim Ali	12345-6000008-4	Male	32	Islam	Intermediate	Employed Full-Time	Shopkeeper/Retail Merchant	Health and Social Work	32000.00
9	Sara Ali	12345-6000009-6	Female	25	Islam	Primary	Student	NULL	NULL	0.00
10	Rashid Hassan	12345-6000010-8	Male	66	Islam	Middle	Employed Full-Time	Crop Farm Laborer	Agriculture and Farming	42000.00
11	Zainab Hassan	12345-6000011-0	Female	60	Islam	Primary	Homemaker	NULL	NULL	NULL
12	Malik Hassan	12345-6000012-2	Male	33	Islam	PHD	Employed Full-Time	Primary School Teacher	Utilities (Water/Electricity)	35000.00
13	Hina Hassan	12345-6000013-4	Female	26	Islam	Matriculation	Student	NULL	NULL	0.00
14	Nasir Hussain	12345-6000014-6	Male	63	Islam	Middle	Employed Full-Time	Construction Laborer	Construction	48000.00
15	Amina Hussain	12345-6000015-8	Female	59	Islam	Uneducated	Homemaker	NULL	NULL	NULL
16	Imran Hussain	12345-6000016-1	Male	30	Islam	Matriculation	Employed Full-Time	Domestic Helper/Maid	Public Administration	28000.00
17	Leila Hussain	12345-6000017-3	Female	22	Islam	Primary	Student	NULL	NULL	0.00
18	Samir Khan	12345-6000018-5	Male	56	Islam	Matriculation	Employed Full-Time	Factory Worker	Manufacturing - Textiles	52000.00
19	Maha Khan	12345-6000019-7	Female	50	Islam	Middle	Homemaker	NULL	NULL	NULL
20	Bashir Khan	12345-6000020-9	Male	28	Islam	Intermediate	Student	Engineer	Wholesale and Retail Trade	0.00
21	Waqar Ahmed	12345-6000021-0	Male	70	Islam	Primary	Unemployed - Not Seeking Work	Software Developer	Financial Services	65000.00
22	Rania Ahmed	12345-6000022-2	Female	66	Islam	Primary	Homemaker	NULL	NULL	NULL
23	Khalid Ahmed	12345-6000023-4	Male	40	Islam	Bachelors	Employed Full-Time	Tailor/Seamstress	Transportation and Logistics	55000.00
24	Dina Ahmed	12345-6000024-6	Female	38	Islam	Intermediate	Employed Full-Time	NULL	NULL	25000.00
25	Jamal Malik	12345-6000025-8	Male	53	Islam	Matriculation	Employed Full-Time	Business Owner	Telecommunications	41000.00
26	Lia Malik	12345-6000026-1	Female	47	Islam	Uneducated	Homemaker	NULL	NULL	NULL
27	Amir Malik	12345-6000027-3	Male	25	Islam	Middle	Student	NULL	NULL	0.00
28	Tariq Hassan	12345-6000028-5	Male	62	Islam	Middle	Employed Full-Time	Agricultural Farmer	Hospitality and Tourism	39000.00
29	Nadia Hassan	12345-6000029-7	Female	57	Islam	Primary	Homemaker	NULL	NULL	NULL
30	Farhan Hassan	12345-6000030-9	Male	32	Islam	Intermediate	Employed Full-Time	Driver/Transport Worker	Education	31000.00
31	Yousef Ali	12345-6000031-0	Male	68	Islam	Primary	Unemployed - Not Seeking Work	Shopkeeper/Retail Merchant	Agriculture and Farming	72000.00
32	Hana Ali	12345-6000032-2	Female	62	Islam	Primary	Homemaker	NULL	NULL	NULL
33	Rasheed Ali	12345-6000033-4	Male	35	Islam	Matriculation	Employed Full-Time	Nurse	Wholesale and Retail Trade	29000.00
34	Saad Khan	12345-6000034-6	Male	60	Islam	Middle	Employed Full-Time	Doctor/Medical Professional	Public Administration	37000.00
35	Luna Khan	12345-6000035-8	Female	54	Islam	Primary	Homemaker	NULL	NULL	NULL
36	Aziz Khan	12345-6000036-1	Male	27	Islam	Middle	Student	NULL	NULL	0.00
37	Adnan Ahmed	12345-6000037-3	Male	64	Islam	Middle	Employed Full-Time	Construction Laborer	Utilities (Water/Electricity)	46000.00
38	Yasmin Ahmed	12345-6000038-5	Female	60	Islam	Uneducated	Homemaker	NULL	NULL	NULL
39	Rana Ahmed	12345-6000039-7	Male	29	Islam	Matriculation	Employed Full-Time	Domestic Helper/Maid	Health and Social Work	24000.00
40	Wasim Malik	12345-6000040-9	Male	56	Islam	Matriculation	Employed Full-Time	Agricultural Farmer	Manufacturing - Textiles	51000.00
41	Rena Malik	12345-6000041-0	Female	51	Islam	Middle	Homemaker	NULL	NULL	NULL
42	Naeem Malik	12345-6000042-2	Male	25	Islam	Primary	Student	NULL	NULL	0.00
43	Mohan Hassan	12345-6000043-4	Male	69	Hinduism	Primary	Employed Full-Time	Software Developer	Agriculture and Farming	58000.00
44	Anjali Hassan	12345-6000044-6	Female	63	Hinduism	Uneducated	Homemaker	NULL	NULL	NULL
45	Rohan Hassan	12345-6000045-8	Male	34	Hinduism	Intermediate	Employed Full-Time	Electrician	Public Administration	26000.00
46	Deepak Ali	12345-6000046-1	Male	62	Hinduism	Middle	Employed Full-Time	Business Owner	Hospitality and Tourism	44000.00
47	Priya Ali	12345-6000047-3	Female	56	Hinduism	Primary	Homemaker	NULL	NULL	NULL
48	Arjun Ali	12345-6000048-5	Male	29	Hinduism	Matriculation	Student	NULL	NULL	0.00
49	Siddiqui Khan	12345-6000049-7	Male	64	Islam	Middle	Employed Full-Time	Crop Farm Laborer	Agriculture and Farming	35000.00
50	Gulnar Khan	12345-6000050-9	Female	59	Islam	Uneducated	Homemaker	NULL	NULL	NULL
51	Farah Khan	12345-6000051-0	Female	33	Islam	Matriculation	Employed Full-Time	NULL	NULL	18000.00
52	Zahir Ahmad	12345-6000052-2	Male	65	Islam	Primary	Employed Full-Time	Driver/Transport Worker	Wholesale and Retail Trade	62000.00
53	Meena Ahmad	12345-6000053-4	Female	60	Islam	Primary	Homemaker	NULL	NULL	NULL
54	Kamil Ahmad	12345-6000054-6	Male	30	Islam	Intermediate	Employed Full-Time	Plumber	Manufacturing - Other	33000.00
55	Ibrahim Raza	12345-6000055-8	Male	66	Islam	Middle	Employed Full-Time	Doctor/Medical Professional	Construction	40000.00
56	Saira Raza	12345-6000056-1	Female	62	Islam	Primary	Homemaker	NULL	NULL	NULL
57	Kashan Raza	12345-6000057-3	Male	30	Islam	Matriculation	Employed Full-Time	Factory Worker	Manufacturing - Textiles	23000.00

Shows the living conditions of each home, tracking things like room counts, clean water access, electricity, and whether it is in a city or a village.

```
mysql>
mysql> SELECT * FROM Household_Living_Standards
    -> WHERE area_classification = 'Rural';
```

household_size	number_of_rooms	census_block_code	area_classification	structure_type	water_supply	lighting_source	cooking_fuel	washroom_type	kitchen_type
4	2	BLK-UC004	Rural	Pure Residential	Motor/Borehole	Electricity Grid	Natural Gas Piping	Flush connected to septic tank	Separate indoor kitchen
4	3	BLK-UC006	Rural	Pure Residential	Tap Water	Electricity Grid	Natural Gas Piping	Flush connected to public sewerage	Separate indoor kitchen
3	4	BLK-UC008	Rural	Mixed Use Residential Commercial	Hand Pump	Solar Power	Firewood	Pit Latrine	Outdoor cooking space
3	3	BLK-UC010	Rural	Mixed Use Residential Commercial	Tap Water	Electricity Grid	Natural Gas Piping	Flush connected to public sewerage	Separate indoor kitchen
3	2	BLK-UC004	Rural	Pure Residential	Hand Pump	Solar Power	Firewood	Pit Latrine	Outdoor cooking space
3	2	BLK-UC016	Rural	Pure Residential	Motor/Borehole	Electricity Grid	LPG Cylinder	Flush connected to septic tank	Separate indoor kitchen
3	3	BLK-UC016	Rural	Commercial Only	Tap Water	Electricity Grid	Natural Gas Piping	Flush connected to public sewerage	No kitchen space

```
7 rows in set (0.00 sec)
```

Groups Pakistan's geographic areas together to show the total number of census blocks and households in every region at a glance.

```
mysql> SELECT province_name, district_name, SUM(gross_estimated_households) AS total_households FROM Regional_Administrative_Summary GROUP BY province_name, district_name ORDER BY total_households DESC;
+-----------------------------+------------------------+------------------+
| province_name               | district_name          | total_households |
+-----------------------------+------------------------+------------------+
| Sindh                       | Karachi Central        |             3440 |
| Sindh                       | Karachi South          |             3350 |
| Islamabad Capital Territory | Islamabad Urban        |             2800 |
| Sindh                       | Karachi East           |             2650 |
| Sindh                       | Hyderabad              |             1910 |
| Balochistan                 | Quetta                 |             1810 |
| Khyber Pakhtunkhwa          | Haripur                |             1730 |
| Khyber Pakhtunkhwa          | Nowshera               |             1710 |
| Khyber Pakhtunkhwa          | Mardan                 |             1660 |
| Khyber Pakhtunkhwa          | Swabi                  |             1600 |
| Khyber Pakhtunkhwa          | Peshawar               |             1530 |
| Khyber Pakhtunkhwa          | Swat                   |             1500 |
| Khyber Pakhtunkhwa          | Abbottabad             |             1450 |
| Sindh                       | Larkana                |             1450 |
| Sindh                       | Shikarpur              |             1440 |
| Sindh                       | Sukkur                 |             1410 |
| Punjab                      | Lahore                 |             1400 |
| Punjab                      | Faisalabad             |             1390 |
| Sindh                       | Jamshoro               |             1370 |
| Gilgit Baltistan            | Gilgit                 |             1350 |
| Punjab                      | Attock                 |             1320 |
| Punjab                      | Jhang                  |             1255 |
| Punjab                      | Multan                 |             1230 |
| Punjab                      | Bahawalpur             |             1215 |
| Punjab                      | Rawalpindi             |             1210 |
| Balochistan                 | Kech (Turbat)          |             1190 |
| Khyber Pakhtunkhwa          | Mansehra               |             1160 |
| Punjab                      | Sheikhupura            |             1100 |
| Sindh                       | Jacobabad              |             1090 |
| Balochistan                 | Khuzdar                |             1080 |
| Khyber Pakhtunkhwa          | Charsadda              |             1080 |
| Sindh                       | Khairpur               |             1080 |
| Islamabad Capital Territory | Islamabad Rural        |             1050 |
| Khyber Pakhtunkhwa          | Chitral                |             1040 |
| Balochistan                 | Kalat                  |             1020 |
| Sindh                       | Dadu                   |             1000 |
| Sindh                       | Ghotki                 |              990 |
| Balochistan                 | Panjgur                |              990 |
| Balochistan                 | Gwadar                 |              950 |
| Punjab                      | Rahim Yar Khan         |              945 |
| Punjab                      | Vehari                 |              935 |
| Punjab                      | Bahawalnagar           |              925 |
| Punjab                      | Khanewal               |              910 |
| Islamabad Capital Territory | Capital Territory East |              900 |
| Punjab                      | Toba Tek Singh         |              875 |
| Khyber Pakhtunkhwa          | Lower Dir              |              870 |
| Balochistan                 | Pishin                 |              870 |
| Balochistan                 | Mastung                |              840 |
| Punjab                      | Kasur                  |              810 |
| Punjab                      | Jhelum                 |              795 |
| Balochistan                 | Chaman                 |              780 |
| Gilgit Baltistan            | Hunza                  |              710 |
| Gilgit Baltistan            | Nagar                  |              660 |
| Khyber Pakhtunkhwa          | Rustam (Proposed)      |              660 |
+-----------------------------+------------------------+------------------+
54 rows in set (0.01 sec)
```

10. Cascading Operations

• ON DELETE CASCADE

In our census architecture, the **Migration_History** table acts as a tracking log for changing districts. It is completely dependent on a person's profile existing in the **Person** table. If a citizen is legally removed from the census database (due to data entry errors or fraud cancellation), keeping their old migration history is meaningless and leaves orphaned logs.

By applying ON DELETE CASCADE, removing a record from the Person table automatically triggers a chain reaction that deletes all corresponding relocation entries from Migration_History.

• ON DELETE SET NULL

Our census operations utilize a supervisor network in the **Census_Staff** table. Field enumerators are monitored via a recursive self-referencing foreign key column called supervisor_id. If a supervisor resigns, retires, or is removed from the staff workforce database, we absolutely **do not** want to lose or delete the subordinate field enumerators they managed.

By using ON DELETE SET NULL, when a supervisor's record is removed, the field enumerators they were managing remain perfectly intact in the database; their supervisor_id simply switches to NULL (meaning they are temporarily unassigned until a new manager takes over).

11. Challenges and Risks

1. Handling Large Data Efficiently

- **The Challenge:** A national census involves capturing records for tens of millions of citizens and millions of households. Standard un-normalized or flat-file database architectures experience catastrophic performance degradation and massive disk space bloating when handling data at this scale.
- **Our Strategy (Normalization & Storage Optimization):**
    - **Elimination of Redundant Text:** Instead of storing repetitive textual data (e.g., storing the string "Tap Water" or "Employed Full-Time" millions of times across individual household and person rows), these attributes were extracted into micro-lookup tables (Water_Source, Employment_Status, etc.).
    - **Strategic Use of Compact Data Types:** By linking parent tracking records via **TINYINT** (1 byte per row, supporting up to 255 distinct lookup categories) instead of INT (4 bytes) or VARCHAR text strings, our schema dramatically compresses the overall database footprint. When scaled to 200 million citizen records, saving 3 bytes per column across dozens of columns prevents gigabytes of unnecessary data footprint, ensuring data blocks fit efficiently into system cache memory during intense analytical processing.
    - **Achieving 3rd Normal Form (3NF):** The database schema was systematically advanced to **3NF**. Every non-prime attribute depends purely, directly, and non-transitively on the primary key anchor of its respective table, ensuring computational efficiency and keeping data storage structurally lean.

2. Ensuring Referential Integrity

- **The Challenge:** In a highly relational database with deep organizational hierarchies (e.g., Province --> Division --> District --> Tehsil --> Union_Council --> Census_Block --> Structure --> Household --> Person), deleting or updating a higher-level record risks creating orphaned child data blocks and breaking internal consistency.
- **Our Strategy:**
  - Implemented structural relational rules via declarative foreign keys.
  - Applied **ON DELETE CASCADE** on strict target operational tracking tables (such as wiping out a citizen's internal logs in Migration_History automatically if their baseline Person profile is deleted).
  - Applied **ON DELETE SET NULL** on organizational management linkages (such as self-referencing workforce hierarchies in Census_Staff), ensuring field enumerator operational performance data is preserved if a supervisor's account is removed.

3. Handling Complex Queries and Optimization

- **The Challenge:** As the volume of deep relational tables grows into millions of data points, executing multi-table statements utilizing numerous INNER JOIN, LEFT JOIN, or nested aggregate operations can lead to slow performance due to intensive nested loop evaluations or extensive full table scans.
- **Our Strategy:**
  - **Relational Views:** Wrapped multi-tier joins inside predefined logical relational SQL views (Most_Populated_Cities, Highly_Educated_Persons, Citizen_Record). This abstracts structural execution complexities and helps the database engine optimize compilation paths.

18. Documentation (10 Marks)

Include a PDF or Word document containing:

- Project Title
- Group Members and Roles
- ERD and Relational Mapping
- SQL Queries with Outputs
- Screenshots
- Explanation of Logic
- Challenges Faced
- Conclusion

19. Video Demonstration & Viva (10 Marks)

Each student must:
- Record a short video showing the project execution
- Explain their assigned role and logic
- Demonstrate SQL queries and outputs
- Prepare for viva questions related to database concepts and implementation

20. Final Submission Must Include

- .sql code files
- Documentation (PDF or Word)
- Screen recording of execution
- ERD and Relational Schema
- Query outputs and screenshots